Jingyi Le

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Education

University of Cambridge — Cambridge, UK
MAST in Mathematics (Part III) — *Oct 2025 – Jun 2026*

- Interests: Number Theory (algebraic, arithmetic, analytic), Pure Dynamics; further in Algebra and Geometry
- Essay Title: Sarnak's Conjecture on Möbius Disjointness (in progress)
- Courses: Local Fields, Analytic Number Field, Algebra in Combinatorics, Commutative Algebra, Algebraic Geometry, Diophantine Analysis, Elliptic Curves

King's College London — London, UK
Bachelor of Science(Hons), Mathematics — *Sep 2022 – May 2025*

- GPA: 1st Class Honor (87); CGPA:89, 3.99/4.00
- Relevant Course:PDE, Number Theory, Real & Complex Analysis, Geometry of Surfaces(1/114), Dynamics, Group, Rings & Modules, Fourier, Galois, Representation Theory, Probability Theory, Numerical

Research Experience

p-adic Dynamics and Failure of Newton's Method — University of Cambridge
Research Intern (Supervisor: Prof. Holly Krieger) — *Jul 2025 – Aug 2025*

- Investigated Faber-Voloch's conjecture on Newton's failures, positive lower density and root bias
- Implemented large-scale computations in Mathematica to test p-adic dynamics and bad reductions
- Provided numerical evidence that apparent root bias vanishes at large prime bounds (still could exist)
- Proved existence of infinitely many primes of convergence and non-convergence in McMullen maps
- Extended research via McMullen and Shub-Smale maps to p-adic convergence, with superattracting fixed points
- Ongoing work; Manuscript in preparation (target: arXiv, Q4 2025)

Hubbard Tree and Entropy in Complex Dynamics — University of Warwick
Research Intern (Supervisors: Dr. Harry Schmidt & Dr. Rob. Kropholler) — *Jul 2024 – Aug 2024*

- Conducted a comprehensive review and proved key theorems on Riemann sphere, Julia sets and Fatou sets
- Modeled Julia Sets and Mandelbrot Sets with Mathematica, applied Newton's Method to quadratic polynomials
- Traced behaviors and structures of dynamics by post-critical finite Hubbard tree and its entropy algorithm
- Discussed progress and upcoming works with two assigned PhD students and the supervisors weekly
- Presented findings at Warwick CDT; outlined potential extensions

Awards & Honors

Cambridge Open Research Fellowship | *awarded to 3 annually by University of Cambridge* — 2025
Mary Lister McCammon Research Fellowship | *awarded by Imperial College London (declined)* — 2025
The University of Hong Kong Summer Research Scholarship | *acceptance rate 5% (declined)* — 2025
Warwick Math UROP Fellowship | *acceptance rate 10%* — 2024

Talks & Presentations

'p-adic Dynamics and Failure of Newton's Method', CMP Day, Cambridge, UK — Aug 2025

Workshop & Conference

PiFORUM25 (Piscopia Initiative Forum), Queen Mary University of London — Sep 2025
Communicating Your Research: Writing Workshop, University of Cambridge — Aug 2025
BICMR Geometric Group Theory Summer School, Peking University — Jun 2025

Skills

Programming: Mathematica/ Python/ R
Typesetting: LaTeX (advanced, for research manuscripts)
Language: Mandarin (native)/ English (fluent)/ Cantonese (conversational)
Research notes, preprints, and diary available at: [github.io]